Practical Digital Filtering Report

Course: Digital Signal Processing (IT00BD97-3001)

Student: Arsenii Marchenko

Part A: Sensor Signal Filtering

1. Experimental Setup & Filter Design

A synthetic sensor signal $x(t)$ of duration $t = 2$ s was generated at a sampling frequency $F_s = 1000$ Hz. The clean signal consists of two low-frequency components (5 Hz and 12 Hz), contaminated by powerline interference (50 Hz), high-frequency noise (137 Hz), and additive Gaussian white noise:

$$x(t) = \sin(2\pi \cdot 5t) + 0.35\sin(2\pi \cdot 12t) + 0.35\sin(2\pi \cdot 50t) + 0.20\sin(2\pi \cdot 137t) + \text{noise}$$

To attenuate the noise above 20 Hz, two low-pass filters were designed with a cutoff frequency $f_c = 20$ Hz:

1. **FIR Filter:** Order 60 (61 taps), designed using fir1 with a Hamming window.
2. **IIR Filter:** 4th-order Butterworth low-pass filter, designed using butter.

Both filters were applied offline using filtfilt to achieve zero-phase distortion and remove intentional phase delays from the performance comparison.

2. Results & Analysis

Time-Domain Performance

(Вставьте сюда График 1: Sensor Time-Domain Comparison)

Over the short 0.2 s observation interval, both the FIR and IIR filtered outputs track the ground-truth clean signal closely. The high-frequency ripples from 50 Hz and 137 Hz interference, as well as the random Gaussian fluctuation, are substantially suppressed.

Frequency Response

(Вставьте сюда График 2: Filter Magnitude Responses)

The magnitude responses computed via freqz show that:

- **FIR (Order 60):** Demonstrates a smooth roll-off with characteristic stopband attenuation and sidelobes typical of window-based design.
- **IIR Butterworth (Order 4):** Provides a sharper transition band and monotonically decreasing magnitude in the stopband with significantly fewer coefficients.

- Both filters pass the -3 dB point precisely at the specified 20 Hz cutoff frequency.

Spectral Content Analysis

(Вставьте сюда График 3: Single-Sided Amplitude Spectra)

Analysis of the single-sided amplitude spectra reveals:

- **Near 5 Hz and 12 Hz:** The desired signals are fully preserved near their nominal amplitudes (1.0 and 0.35 respectively).
- **Near 50 Hz and 137 Hz:** Both filters attenuate the powerline interference (50 Hz) and high-frequency disturbance (137 Hz) down close to the noise floor (> 30 dB attenuation).

Quantitative RMSE Comparison

Signal State	Root Mean Square Error (RMSE)
Noisy Input (x)	≈ 0.3802
FIR Output (y_{FIR})	≈ 0.0504
IIR Output (y_{IIR})	≈ 0.0452

- **RMSE Interpretation:** RMSE quantifies the total energy difference between the filtered output and the known clean signal. While it confirms significant overall noise reduction (an ~87% drop in error), RMSE cannot distinguish between residual high-frequency noise, amplitude attenuation at target frequencies, or edge-transient artifacts.

3. FIR vs. IIR Structural Comparison

- **Coefficient/Order Count:** The FIR filter requires 61 coefficients, whereas the IIR filter requires only 9 coefficients (5 numerator b, 4 denominator a).
- **Transition Behavior:** The 4th-order Butterworth filter achieves a steeper roll-off in the transition region (20 Hz $-$ 50 Hz) than the order-60 FIR filter.
- **Edge Effects:** Minor transient distortion occurs at the boundaries ($t \approx 0$ s and $t \approx 2$ s) due to initial condition padding in filtfilt.
- **Preferred Use Cases:**

- **FIR:** Preferred in audio and communications where strict linear-phase response and guaranteed absolute stability are required.
- **IIR:** Preferred in resource-constrained embedded systems (e.g., microcontrollers) where low computational complexity and minimal memory consumption are vital.

Part B: Audio Hum Reduction

1. Setup & Notch Filter Implementation

The built-in handel audio dataset ($F_s = 8192$ Hz) was corrupted with a synthetic 50 Hz sinusoidal hum at an amplitude of 0.15.

A 2nd-order IIR Butterworth band-stop (notch) filter was designed with a narrow stopband (49 Hz $- 51$ Hz) and applied via filtfilt.

2. Results & Observations

Waveform and Spectrogram Analysis

(Вставьте сюда Графики 4 и 5: Audio Waveforms & Spectrograms)

- **Waveform:** The continuous low-frequency sinusoid super-imposed on the time-domain signal is visually eliminated after notch filtering.
- **Spectrogram:** In the corrupted signal spectrogram, a distinct horizontal band appears continuously at 50 Hz. After applying the narrow band-stop filter, this line disappears completely without degrading adjacent spectral content.
- **Listening Test:** Auditory playback confirms that the annoying 50 Hz hum is fully eliminated while preserving the overall tonal quality and clarity of the original audio.

Stopband Width Comparison ($49 - 51$ Hz vs. $45 - 55$ Hz)

Expanding the stopband width to $45 - 55$ Hz provides greater tolerance if the powerline frequency drifts slightly. However, the wider notch removes a broader range of low-frequency audio energy, which can cause audible loss of warmth and bass fidelity in musical signals.

Report Answers to Technical Questions

1. Why is a 20 Hz low-pass reasonable for Part A, and what assumption does it make about the wanted sensor signal?

A 20 Hz cutoff frequency is chosen because the desired signal components lie at 5 Hz and 12 Hz, well below 20 Hz, while the lowest interference source (50 Hz) is well above

it. This selection assumes that the true sensor signal is strictly bandlimited to < 20 Hz and contains no useful high-frequency transients or fast step changes above this limit.

2. Which filter gives the better result for the requirements you chose? Support the answer with at least two kinds of evidence.

The 4th-order Butterworth **IIR filter** gives the superior overall performance for this offline task:

1. **Numerical Evidence:** The IIR filter achieves a lower RMSE (≈ 0.0452) compared to the FIR filter (≈ 0.0504).
2. **Computational Evidence:** The IIR filter achieves superior stopband attenuation using only 9 coefficients, compared to 61 coefficients for the FIR filter, making it over $6\times$ more computationally efficient.

3. How would your implementation change if the sensor had to be filtered live with low latency?

For a real-time, low-latency implementation:

- filtfilt **cannot be used** because it requires non-causal access to future samples.
- Filtering would be performed using causal filter() or a direct difference equation loop.
- An **FIR filter** with linear phase (constant group delay) would be preferred to avoid non-linear phase distortion across signal frequencies, and the order would be kept low to minimize processing latency.

4. Why is a narrow band-stop appropriate for a steady 50 Hz hum? What would you do if the interference frequency drifted?

A narrow notch filter is ideal for a stationary 50 Hz hum because it removes the interference while preserving almost all surrounding audio frequencies. If the frequency drifts (e.g., 48 Hz $-$ 52 Hz due to power grid load changes), a static narrow filter will fail. Solutions include:

- Using an **Adaptive Notch Filter** (e.g., LMS/RLS algorithm) that dynamically tracks the hum frequency.
- Widening the stopband, accepting the trade-off of minor audio signal loss.

5. Describe one limitation of your experiments and one improvement you would make in a real measurement system.

- **Limitation:** The noise and interference added were synthetic, stationary, and deterministic. Real-world sensor signals suffer from non-stationary noise, powerline harmonics (100 Hz, 150 Hz), and dynamic baseline wander.

- **Improvement:** Implement analog anti-aliasing hardware filtering prior to the ADC stage, combine it with oversampling, and utilize adaptive digital filtering to handle non-stationary real-world noise environments.